

Food Delivery Delay Analysis and Performance Insights

Course: C-DE211 – Data Analysis
Institution: Egypt University in Egypt (EUI)

Prepared by: Lucinda Ahmed 24-101273

Farida Waleed 24-101081

Farah Mohamed 24-101171

Kenzy Essam 24-101065

SECTION 1 - ABSTRACT:

This project analyzes food delivery patterns and delivery delays using a dataset of 20,000 orders obtained from Kaggle. The study investigates how various factors such as traffic conditions, weather conditions, delivery method, and delivery distance influence delivery delays and customer satisfaction. Exploratory Data Analysis (EDA) techniques were applied to identify trends and patterns in ordering behavior across months and days of the week. In addition, visualization tools including heatmaps and comparative plots were used to highlight relationships between variables.

To further validate the findings, hypothesis testing was conducted using ANOVA to determine whether factors such as traffic conditions significantly affect delivery delays. The results indicate that delivery distance and traffic conditions play an important role in influencing delays, while weather conditions have a moderate effect. However, it is important to note that the dataset appears to be simulated, as some expected real-world relationships—such as consistently higher delays under high traffic conditions—are not strongly reflected.

Overall, the analysis provides insights into delivery performance and highlights potential areas for optimization to improve efficiency and customer satisfaction.

SECTION 2 - INTRODUCTION:

In recent years, food delivery services have experienced significant growth, becoming an essential part of everyday life. With the rise of online platforms and mobile applications, customers increasingly rely on fast and reliable delivery services. As a result, delivery efficiency and timeliness have become critical factors influencing customer satisfaction and business success.

One of the main challenges faced by delivery services is managing delays caused by various factors such as traffic congestion, weather conditions, and delivery

distance. Understanding these factors is essential for optimizing operations, reducing delays, and improving the overall customer experience.

This project aims to analyze food delivery data to identify patterns in delivery performance and determine the key factors that influence delivery delays. By applying data analysis techniques and statistical testing, the study seeks to provide insights that can help improve delivery efficiency and decision-making in logistics systems.

SECTION 3 - DATASET DESCRIPTION:

The dataset used in this project was obtained from Kaggle and contains information about food delivery orders. It consists of 20,000 records and 31 columns, providing a comprehensive overview of delivery-related variables.

Key features in the dataset include:

- Order time (date)
- Delivery delay
- Traffic condition (Low, Medium, High)
- Weather condition (Sunny, Rainy, Snowy)
- Delivery method (Bike, Car, Walk)
- Customer satisfaction
- Delivery distance
- Order value

It is important to note that the dataset appears to be synthetically generated and may not fully reflect real-world relationships. For example, high traffic conditions do not consistently result in higher delivery delays, suggesting that the data lacks strong causal relationships between variables. This limitation is considered when interpreting the results of the analysis.

SECTION 4 - METHODOLOGY :

4.1 Data Preparation

The dataset was examined for quality and consistency before analysis. Since there were no missing or inconsistent values, no data cleaning or preprocessing steps were required. All variables were used directly in the analysis.

4.2 Exploratory Data Analysis (EDA)

EDA was conducted to understand the distribution and behavior of key variables, these steps helped identify trends and potential patterns in the data.

4.3 Data Visualization

Various visualization techniques were used to better understand relationships between variables. Bar charts and comparative plots were used to analyze categorical variables such as traffic condition, weather condition, and delivery method.

4.4 Analysis of Influencing Factors

The project examined how different factors affect delivery delay. These factors include:

- Traffic conditions (Low, Medium, High)
- Weather conditions (Sunny, Rainy, Snowy)
- Delivery method (Bike, Car, Walk)
- Delivery distance

Each factor was analyzed individually to observe its impact on delivery performance.

4.5 Hypothesis Testing

To statistically validate the observed patterns, hypothesis testing was performed using Analysis of Variance (ANOVA). This test was used to determine whether there are significant differences in delivery delays across different traffic conditions.

The hypotheses were defined as follows:

- Null Hypothesis (H0): Traffic condition has no significant effect on delivery delay.

- Alternative Hypothesis (H1): Traffic condition has a significant effect on delivery delay.

A significance level of 0.05 was used to evaluate the results. Based on the p-value obtained from the ANOVA test, a decision was made to either reject or fail to reject the null hypothesis.

SECTION 5 - RESULTS & DISCUSSION :

5.1 Delivery Delay Distribution

The analysis of delivery delays shows that most orders fall within a moderate delay range, with fewer instances of extremely high or very low delays.

5.2 Order Patterns Over Time

The distribution of orders across months and days of the week reveals certain periods show higher order volumes, indicating peak demand times so it highlights the importance of resource planning to handle increased workload during busy periods.

5.3 Impact of Traffic Conditions

Some variation in delay was observed across low, medium, and high traffic conditions, the relationship was not consistently strong. This inconsistency suggests that the dataset may not fully capture realistic traffic impacts.

5.4 Impact of Weather Conditions

Orders during rainy or snowy conditions tended to experience slightly higher delays compared to sunny conditions.

5.5 Delivery Method Analysis

Faster delivery methods generally resulted in lower delays, while slower methods such as walking tended to have higher delays.

5.6 Delivery Distance

Delivery distance was one of the most influential factors affecting delay. As distance increased, delivery delays also tended to increase.

5.7 Customer Satisfaction

Higher delivery delays generally corresponded to lower satisfaction levels. This emphasizes the importance of minimizing delays to maintain a positive customer experience.

5.8 Hypothesis Testing Results

The null hypothesis was rejected, suggesting that traffic conditions do have an effect on delivery delays.

SECTION 6 - CONCLUSION

This project analyzed food delivery data to identify patterns and factors influencing delivery delays. The results show that delivery distance is the most significant factor affecting delays, while traffic and weather conditions also contribute to variations in delivery performance.

Overall, the findings highlight the importance of optimizing delivery operations by considering distance, environmental conditions, and delivery methods. Improving these factors can lead to reduced delays and higher customer satisfaction.

SECTION 7 — LIMITATIONS:

While this project provides useful insights into food delivery patterns, several limitations should be considered when interpreting the results.

First, the dataset appears to be synthetically generated and may not fully represent real-world delivery operations. Some expected relationships—such as consistently higher delivery delays under high traffic conditions—are not strongly reflected in the data. This suggests that the dataset may lack realistic causal relationships between variables.

Second, the order time variable includes only the date and does not provide the exact time of the order.

Finally, since the dataset does not represent a specific geographic location, the findings cannot be directly generalized to real delivery systems in Egypt or other regions.

SECTION 8 - RECOMMENDATIONS

First, delivery services should prioritize route optimization to reduce travel distance and minimize delays. Using intelligent routing systems can help drivers avoid inefficient paths and reduce delivery time.

Second, delivery methods should be selected based on distance and environmental conditions. For example, bikes may be more efficient for short distances, while cars may be better suited for longer routes.

Finally, future analyses should use real-world datasets with more detailed variables, such as exact delivery times and geographic information

SECTION 9 - REFERENCES

Kaggle Dataset: Food Delivery Dataset

<https://www.kaggle.com/datasets/varshinipallerla/food-delivery>

Python Libraries:

Pandas, NumPy, Matplotlib, Seaborn, Scipy.